

Linear Programming for Transportation and Logistics Optimization: Background and Related Works

COMP 6704 Group Project

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1 Background and Preliminaries

1.1 The Transportation Problem as a Linear Program

The linear programming (LP) model traces its origins to production planning [16], and its first large-scale application was the classical transportation problem [9, 11]. Given a set of m supply nodes (e.g., depots) with supplies $s_i \geq 0$, a set of n demand nodes (e.g., retailers) with demands $d_j \geq 0$, and per-unit shipping costs $c_{ij} \geq 0$, the *transportation problem* asks for a shipping plan $x \in \mathbb{R}_{\geq 0}^{mn}$ that minimizes total cost:

$$\min_x \quad \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \tag{1}$$

$$\text{s.t.} \quad \sum_{j=1}^n x_{ij} \leq s_i, \quad i = 1, \dots, m, \tag{2}$$

$$\sum_{i=1}^m x_{ij} \geq d_j, \quad j = 1, \dots, n, \tag{3}$$

$$x_{ij} \geq 0. \tag{4}$$

Constraints (2) encode depot capacity and constraints (3) encode demand satisfaction. The model is intentionally parsimonious—its objective and constraints are all linear in the decision variables—because unit transportation costs are, to a good approximation, proportional to shipped quantities, and because this linearity is what makes an exact, globally optimal solution computable at industrial scale. Richer logistics settings retain this LP core: multi-commodity variants stack one such program per commodity and couple them through shared arc capacities [8, 29], vehicle-routing formulations relax it through linear cut constraints [22], and fixed-charge or multimodal extensions build directly on its linear relaxation [26, 27].

1.2 Duality and Economic Interpretation

Associating dual variables $u_i \leq 0$ with the supply constraints (2) and $v_j \geq 0$ with the demand constraints (3), the dual of (1)–(3) is

$$\begin{aligned} \max_{u,v} \quad & \sum_{j=1}^n d_j v_j - \sum_{i=1}^m s_i u_i \\ \text{s.t.} \quad & v_j - u_i \leq c_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n. \end{aligned} \tag{5}$$

By strong duality, the optimal values of (1) and (5) coincide, and the complementary slackness conditions

$$(v_j - u_i - c_{ij}) x_{ij} = 0 \quad \forall i, j \quad (6)$$

characterize optimality. In logistics terms, the dual variables are *shadow prices*: u_i is the marginal value of one additional unit of capacity at depot i , and v_j is the marginal cost of serving one additional unit of demand at retailer j . This interpretation is a principal reason exact LP methodology—as opposed to heuristics—is considered essential in transportation planning: the optimal basis simultaneously certifies a shipping plan and a consistent system of marginal prices for capacity contracting and what-if analysis.

1.3 Structure of the Transportation Polytope

The feasible region of (1)–(3) is the *transportation polytope*. Three structural properties govern algorithmic behavior on it. First, its constraint matrix is totally unimodular, so when supplies and demands are integral every vertex is integral: the LP relaxation is exact and no integer-programming machinery is required. Second, when total supply equals total demand, constraints (2)–(3) are linearly dependent (any one constraint is implied by the rest), so every basic feasible solution has at most $m + n - 1$ positive shipments; this rank deficiency makes the polytope *prone to degeneracy*, a property examined in detail by Im and Wolkowicz [15]. Third, the constraint matrix is sparse—each variable x_{ij} appears in exactly two constraints—so the polytope is an instance of a network-flow polytope, and algorithms that exploit graph structure (Section 2.4) can act directly on the underlying bipartite graph rather than on the mn -column tableau.

These properties jointly define the computational landscape surveyed in Section 2: integrality guarantees exactness of the LP core; degeneracy stresses simplex-type implementations; and network sparsity is the lever that structure-exploiting algorithms pull.

2 Related Works

Research on solving transportation and logistics LPs can be organized into five methodological families: simplex-type algorithms (§2.1), interior-point methods (§2.2), first-order methods and GPU-accelerated solvers (§2.3), structure-exploiting and decomposition approaches (§2.4), and learning-augmented optimization (§2.5). We survey each family in turn and conclude with the bottlenecks that delineate the current research frontier (§2.8).

2.1 Simplex-Type Methods

Dantzig’s simplex method [9] remains the algorithmic backbone of transportation planning. The revised and dual revised variants dominate modern implementations because they compress the tableau to a basis and permit sparse linear algebra; Huangfu and Hall [13] study the parallelization of the dual revised simplex method and identify pricing and basis update as the only phases amenable to concurrent execution. For transportation and network-flow problems specifically, the *network simplex* method exploits the spanning-tree structure of optimal bases, replacing dense basis factorization with graph operations; implementations and extensions are studied for maximum flow [30] and for budget-constrained minimum cost flow [12]. On the theoretical side, the worst-case exponential behavior of the simplex method is not representative of practical instances: Cornelissen and Manthey [7] establish smoothed polynomial bounds for the network simplex and the minimum-mean cycle canceling algorithm, providing the most convincing available explanation of its empirical

efficiency. Industrial solvers such as HiGHS [14] still default to the dual simplex as the first algorithm attempted on LPs.

2.2 Interior-Point Methods

Interior-point methods (IPMs), initiated by Karmarkar [17], attain polynomial worst-case complexity by following the central path. Accessible derivations of the primal-dual path-following scheme are given by Mehlhorn and Saxena [21]. Two research threads address the practical weaknesses of IPMs on large instances. First, the Newton system arising at each iteration becomes increasingly ill-conditioned as the iterate approaches the optimal face; Cornelis and Vanroose [6] prove convergence of a regularized inexact IPM in which the Newton equations are solved only approximately with a regularization term stabilizing the linear algebra. Second, a sequence of works seeks to remove the expensive direct factorization step: Lin et al. [19] propose an ADMM-based IPM in which the augmented system is handled by operator splitting rather than sparse Cholesky, trading per-iteration exactness for scalability. Rothberg [25] revisits this trade-off from the opposite direction, showing that a first-order method can provide a warm-started, stabilized interior-point phase.

2.3 First-Order Methods and GPU-Accelerated Solvers

The most active recent line of work targets LP instances far beyond the reach of factorization-based methods. PDL [1] established primal-dual hybrid gradient (PDHG) with restart heuristics as a practical first-order LP solver at the million-variable scale; cuPDL.x [20] demonstrated that a GPU implementation with preconditioning and scaling reaches solutions competitive with, and orders of magnitude faster than, commercial simplex and IPM codes on huge instances. Subsequent work scales the paradigm further along three axes: distribution across multi-GPU clusters [18], presolving tailored to the numerical behavior of first-order iterations [5], and automatic parameter tuning with generalization guarantees [24]. This family trades exactness for parallelizability: solutions are certified only to a moderate tolerance, and tail convergence is slow.

2.4 Structure-Exploiting and Decomposition Methods

A complementary philosophy is to exploit the combinatorial or decomposable structure of logistics LPs rather than to accelerate generic solvers. For multi-commodity flow—the natural multi-depot generalization of the transportation problem—Dai et al. [8] develop efficient LP formulations for the splitted variant, while van den Brand and Zhang [29] obtain high-accuracy solutions by reduction to single-commodity subproblems. At the industrial scale of end-to-end supply chain planning, monolithic LP/IPP formulations become intractable and decomposition is indispensable: Guan et al. [10] combine column generation with Benders decomposition to solve enterprise-scale paper-industry planning, with the pricing subproblems constituting the computational crux. Relaxations that depart from pure network structure, such as fixed-charge network flow, require LP-based support identification to remain tractable [26].

2.5 Learning-Augmented Optimization

A nascent direction applies machine learning not to replace exact solvers but to accelerate their inner loops. Sul et al. [28] train a reinforcement learning policy to select PDHG step sizes and restart decisions, accelerating first-order LP solving without sacrificing feasibility certificates, and Prasad and Sharma [24] provide learning-theoretic guarantees for tuning GPU solver parameters on a distribution of instances. The common caveat is generalization: learned accelerators are only

guaranteed on distributions resembling the training instances, which motivates the hybridization trend of §2.2–§2.3 as a more conservative alternative.

2.6 Problem-Specific Theory for Transportation

Beyond generic solver technology, the transportation problem itself continues to motivate specialized algorithms and theory. Baretto et al. [3] propose Iterated Inside Out, an exact algorithm that outperforms general-purpose LP solvers on classical transportation instances. Nickel et al. [23] characterize the problem under Monge cost structure, obtaining closed-form solutions that transfer to discrete ordered median problems. On the applied side, Bai [2] compare Sinkhorn-type entropy-regularized schemes against exact LP solvers for partial optimal transport, quantifying the accuracy gap incurred by regularization. Finally, Im and Wolkowicz [15] develop the theory of degeneracy, strict feasibility, and stability in LP—directly relevant because transportation polytopes are a canonical source of highly degenerate instances.

2.7 Applications in Transport and Logistics

At the modeling level, Bridgelall [4] provide a tutorial treatment of LP formulations across supply chain and transport logistics, and Munari et al. [22] give a generalized formulation for vehicle routing that subsumes many LP relaxations in the literature. Su et al. [27] integrate buses and drones in a multimodal freight model for last-mile delivery, illustrating how the linear LP core is embedded in richer logistics decision problems.

2.8 Bottlenecks and Open Gaps

Three tensions emerge from this literature and jointly define the frontier. *Accuracy versus scale*: simplex and IPM deliver machine-precision, vertex-certified solutions whose bases support the shadow-price analysis of Section 1.2, but both are limited by factorization-based linear algebra; first-order methods scale to extreme sizes and parallelize naturally, yet terminate at moderate accuracies with slow tail convergence [1, 20]. *Genericity versus structure*: structure-exploiting algorithms are dramatically faster on pure network LPs but fragile once side constraints or fixed charges appear [12, 26]. *Speed versus reliability in learning-augmented solvers*: learned components accelerate but lack distributional guarantees [24, 28]. The prevailing response is hybridization—combining first-order and interior-point phases [19, 25]—while degeneracy, endemic to transportation polytopes, remains a stressor for all vertex-traversing methods [15]. These observations directly motivate the experimental design of this project: transportation instances simultaneously exercise simplex degeneracy handling, IPM conditioning, and network structure, making them an informative benchmark for cross-family algorithm comparison.

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